

Redrock Study Simulation

Aldarin Island: Dashed Coyotes. Printed from the same content files the simulation reads, so it cannot drift out of step with what a candidate is marked against. For WK and Francesco; nothing in the simulation links here.

29

Investigation items

8

Analysis boxes

13

Report parts

6

Cases

56

Total marks

Each Investigation item earns a full mark if it was collected before the candidate first left the Investigation. One fetched on a return visit from the Analysis earns half a mark (set by `return_visit_weight` in `version.json`, currently `0.5`). One never collected earns nothing.

Investigation

29 pieces of information to collect

Research Information

#	VALUE	LABEL	NEEDED FOR
1	Review the data and identify the percentage of prey death in Mistveil and Bluebell Woods due to Dashed Coyotes.	Objective	Every question — it defines which two zones matter

Exhibit 1: Primary Prey Population Across Zones on Aldarin Island

#	VALUE	LABEL	NEEDED FOR
2	an insignificant portion of their diets	Other preys' contribution towards Dashed Coyote diets	Question 2 — the four prey make up all kills
3	550	Marmot population in Mistveil Woods	Question 3
4	820	Chinchilla population in Mistveil Woods	Question 3
5	1,200	Hare population in Mistveil Woods	Question 3
6	430	Gopher population in Mistveil Woods	Question 3
7	700	Marmot population in Bluebell Woods	Question 3
8	1,080	Chinchilla population in Bluebell Woods	Question 3
9	1,360	Hare population in Bluebell Woods	Question 3
10	610	Gopher population in Bluebell Woods	Question 3

Exhibit 2: Dashed Coyote Hunting Behavior Across Zones on Aldarin Island

#	VALUE	LABEL	NEEDED FOR
11	number of the four mapped prey killed per day per Coyote	Definition of kill rate	Question 1

#	VALUE	LABEL	NEEDED FOR
12	the four species are killed proportionately to their population	Dashed Coyote hunting behavior	Questions 3 and 4
13	Once calculated, the kill rate is expected to remain constant throughout the year (365 days)	Kill rate over the year	Question 2
14	additional 8 days	Additional Dashed Coyote tracking time	Question 1
15	Mistveil Woods pack size was undercounted by 1	Additional data #3	Question 1
16	Mistveil Woods prey killed increased by 30	Additional data #4	Question 1
17	Bluebell Woods prey killed increased by 26	Additional data #5	Question 1
18	0.35	Kill rate in Mistveil Woods	Question 1 (and the Report's kill-rate comparison)
19	30	Total prey killed in Mistveil Woods	Question 1 (and the Report's kill-rate comparison)
20	17	Days tracked in Mistveil Woods	Question 1 (and the Report's kill-rate comparison)
21	5	Pack size in Mistveil Woods	Question 1 (and the Report's kill-rate comparison)
22	0.30	Kill rate in Bluebell Woods	Question 1 (and the Report's kill-rate comparison)
23	40	Total prey killed in Bluebell Woods	Question 1 (and the Report's kill-rate comparison)
24	12	Days tracked in Bluebell Woods	Question 1 (and the Report's kill-rate comparison)
25	11	Pack size in Bluebell Woods	Question 1 (and the Report's kill-rate comparison)

Exhibit 3: Estimated Death Rate from Illness and Disease

#	VALUE	LABEL	NEEDED FOR
26	10	Marmot death rate due to illness and disease (%)	Question 3
27	15	Chinchilla death rate due to illness and disease (%)	Question 3
28	5	Hare death rate due to illness and disease (%)	Question 3
29	10	Gopher death rate due to illness and disease (%)	Question 3

Only the two zones named in the objective — Mistveil Woods and Bluebell Woods — matter. Everything about Teklan Hills and Rosefield is a distractor, and so is the introduction date of the coyotes.

Analysis

8 answer boxes

Q	BOX	ANSWER	ACCEPTED
	Question 1 What is the new kill rate for Dashed Coyotes in Mistveil and Bluebell Woods given the additional data. Round your answer to the closest 1 decimal point.		

Q	BOX	ANSWER	ACCEPTED
Q1	Mistveil Woods	0.4	0.35 to under 0.45 · 1 decimal place From Exhibit 2, Mistveil Woods had 30 prey killed over 17 days by a pack of 5. With the additional data: 60 killed (extra 30), 25 days (extra 8), pack of 6 (undercounted by 1). New kill rate = total prey killed ÷ (days tracked × pack size) = $60 \div (25 \times 6) = 60 \div 150 = 0.4$.
Q1	Bluebell Woods	0.3	0.25 to under 0.35 · 1 decimal place From Exhibit 2, Bluebell Woods had 40 prey killed over 12 days by a pack of 11. With the additional data: 66 killed (extra 26), 20 days (extra 8), pack unchanged at 11. New kill rate = $66 \div (20 \times 11) = 66 \div 220 = 0.3$.
Question 2 What is the total number of prey killed per year (365 days) by Dashed Coyotes in Mistveil and Bluebell Woods? Assume that the four primary prey make up all of Dashed Coyotes kills. Round up your answer to the closest whole number.			
Q2	Mistveil Woods	876	875.5 to under 876.5 · 0 decimal places Prey killed per year = new kill rate × 365 days × new pack size = $0.4 \times 365 \times 6 = 876$.
Q2	Bluebell Woods	1205	1204.5 to under 1205.5 · 0 decimal places $0.3 \times 365 \times 11 = 1,204.5$, which rounds up to 1,205.
Question 3 What is the average death rate due to illness and disease, taking into considering all four primary prey, in Mistveil and Bluebell Woods? Round your answer to the closest whole percentage point.			
Q3	Mistveil Woods	9%	8.5 to under 9.5 · 0 decimal places Total prey in Mistveil Woods = $550 + 820 + 1,200 + 430 = 3,000$. Deaths from illness = $550 \times 10\% + 820 \times 15\% + 1,200 \times 5\% + 430 \times 10\% = 55 + 123 + 60 + 43 = 281$. Weighted death rate = $281 \div 3,000 = 0.094 \approx 9\%$.
Q3	Bluebell Woods	10%	9.5 to under 10.5 · 0 decimal places Total prey in Bluebell Woods = $700 + 1,080 + 1,360 + 610 = 3,750$. Deaths from illness = $70 + 162 + 68 + 61 = 361$. Weighted death rate = $361 \div 3,750 = 0.096 \approx 10\%$.
Question 4 What percentage of yearly prey lost in the Mistveil and Bluebell Woods are due to Dashed Coyotes? Round your answer to the closest whole percentage point.			
Q4	Mistveil Woods	76%	75.5 to under 76.5 · 0 decimal places Total deaths in Mistveil Woods = coyote kills 876 + illness deaths 281 = 1,157. Share due to coyotes = $876 \div 1,157 = 0.757 \approx 76\%$.
Q4	Bluebell Woods	77%	76.5 to under 77.5 · 0 decimal places Total deaths in Bluebell Woods = $1,205 + 361 = 1,566$. Share due to coyotes = $1,205 \div 1,566 = 0.769 \approx 77\%$.

Report

13 parts

Written

With the additional data, the kill rate for Dashed Coyotes in Mistveil Woods **A: will** increase. Under the new rate, annual prey killed is expected to be **B: 876** compared to disease deaths of **C: 281**. **D: 76** % of the total deaths were due to the Dashed Coyotes in Mistveil Woods.

With the additional data, the kill rate for Dashed Coyotes in Bluebell Woods **E: won't** increase. Under the new rate, annual prey killed is expected to be **F: 1205** compared to disease deaths of **G: 361**. **H: 77** % of the total deaths were due to the Dashed Coyotes in Bluebell Woods.

BLANK	ANSWER	KIND	ACCEPTED
A	will	choice	that option only Mistveil's new kill rate (0.4) is higher than the old one in Exhibit 2 (0.35), so it will increase.
B	876	number	875.5 to under 876.5 · 0 decimal places Annual prey killed in Mistveil Woods, from Question 2.
C	281	number	280.5 to under 281.5; or 269.5 to under 270.5 (270 also accepted) · 0 decimal places Illness deaths in Mistveil Woods, from the working of Question 3 (281). A candidate who applied the rounded 9% to 3,000 gets 270, which is also accepted.
D	76	number	75.5 to under 76.5 · 0 decimal places From Question 4.
E	won't	choice	that option only Bluebell's new kill rate (0.3) equals the old one (0.30), so it won't increase.
F	1205	number	1204.5 to under 1205.5 · 0 decimal places Annual prey killed in Bluebell Woods, from Question 2.
G	361	number	360.5 to under 361.5; or 374.5 to under 375.5 (375 also accepted) · 0 decimal places Illness deaths in Bluebell Woods, from the working of Question 3 (361). Applying the rounded 10% to 3,750 gives 375, also accepted.
H	77	number	76.5 to under 77.5 · 0 decimal places From Question 4.

Graph

PROMPT	ANSWER	WHY
Select the best graph to visualize how the total deaths across Mistveil and Bluebell Woods break down by individual zones and causes.	PIE	The pie chart shows the components of a single variable; the line chart shows trends over time; the bar chart shows the distribution of a dataset at a given time. We want to break total deaths down by zone and cause, so the pie chart is the most appropriate.

Visual: the figures

Add the relevant information to show how the total deaths across Mistveil and Bluebell Woods break down by individual zones and causes.

	MISTVEIL WOODS	BLUEBELL WOODS
Deaths due to Dashed Coyotes	876 875.5 to under 876.5 · 0 decimal places	1205 1204.5 to under 1205.5 · 0 decimal places
Deaths due to Illness and Disease	281 280.5 to under 281.5; or 269.5 to under 270.5 (270 also accepted) · 0 decimal places	361 360.5 to under 361.5; or 374.5 to under 375.5 (375 also accepted) · 0 decimal places

Deaths due to Dashed Coyotes come from Question 2 (876 and 1,205); deaths due to illness and disease from the working of Question 3 (281 and 361).

CASE	QUESTION	KIND	ANSWER
1	Assuming: • P represents future population • n represents number of years from present • The population will double every 7 years • The population shown above represent the initial population at present What is the best equation to calculate the future Horned Armadillo population? Only two formula shapes are offered: $\text{constant} \times 2^{(n/7)}$, and $(\text{constant} \div n) + 2^7$. Doubling every 7 years is the first shape, so the answer is b or d. The constant must be the present population: $310 + 230 + 160 + 120 = 820$. So $P = 820 \times 2^{(n/7)}$.	choose one	$P = 820 \times 2^{(n/7)}$
2	What information is used to determine the new total number of kills in Teklan Hills over the coming year? Collect the relevant data into the Research Journal. Only Teklan Hills matters, so the Rosefield column of both tables can be ignored; and only the coming year matters, so the growth rows can be ignored. Total kills = kills from existing packs + kills from new packs, and each = number of packs $\times$ annual hunts per pack $\times$ kills per hunt. So the six items needed are, for Teklan Hills in both tables: number of packs, average annual hunts per pack, and average kills per hunt. Pack size is not needed.	collect	# of existing packs in Teklan Hills: 3 Average annual hunts per existing pack in Teklan Hills: 40 Average kills per hunt for existing wolves in Teklan Hills: 3 # of new packs in Teklan Hills: 1 Average annual hunts per new pack in Teklan Hills: 42 Average kills per hunt for new wolves in Teklan Hills: 4
3	What is the expected number of new infections for Month 12? Round up your answer to the closest whole number. Growth from Month 1 to Month 6 = $225 \div 60 - 1 = 2.75 = 275\%$. Growth from Month 6 to Month 12 = $275\% + 15 \text{ points} = 290\%$. Month 12 infections = $225 \times (1 + 2.9) = 877.5$, rounded up to 878.	number	Answer = 878 (877.5 to under 878.5, 0 decimal places)
4	Which chart type best visualizes how the new infections evolved over time across the four zones of the Aldarin Island? A line graph is the best choice for trends over time. A bar chart shows the distribution of a dataset at a given time; a scatterplot shows the correlation between two variables; a pie chart shows the components of a single variable.	dropdowns	Answer: Line Graph
5	Which of the following statements are supported by the data? You may select more than one. Only statement b can be read directly from the table. Do not select a statement unless the data makes it certain.	choose many	Reef Petrels had their strongest breeding improvement in West Shoals.
6	Report the mean, mode, and median weekly sightings based on the data above. With 20 values it is safer to total each coast first, then combine.	numbers	Mean = 21.3 (21.25 to under 21.35, 1 decimal place) Median = 21 (20.5 to under 21.5, 0 decimal places) Mode = 15 (14.5 to under 15.5, 0 decimal places)

Weighted score out of 100 = each phase's score divided by its best possible, times its weight, added up. The percentile is read off the table in a straight line between its points, rounded to a whole number and kept between 1 and 99. Results mode: `full`.

Phase weights

PHASE	WEIGHT
Investigation	25
Analysis	25
Report	25
Cases	25

Zones

PERCENTILE	ZONE
0 to 69	Below 70th
70 to 79	Borderline
80 to 89	Likely pass
90 and above	Comfortable

Table: weighted score, then the percentile it reads as

SCORE	PERCENTILE	SCORE	PERCENTILE	SCORE	PERCENTILE	SCORE	PERCENTILE
0	1	20	5	35	12	45	20
55	30	62	40	68	50	74	60
79	68	84	75	88	80	91	85
94	90	96	94	98	97	100	99

Footnote shown to the candidate: This percentile is our own estimate for this simulation, not a McKinsey figure. McKinsey does not publish its exact scoring, so we weight each section equally for the purpose of scoring. Candidates who pass typically score in the top quartile, but we recommend aiming for the 90th and above.